

PORTAGE GLACIER, AK, USA

WMO: 700001

Lat: 60.785N

Lon: 148.841W

Elev: 31

StdP: 100.95

Time Zone: -9.00 (NAK)

Period: 04-19

WBAN: 26492

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.1	-21.8	-27.9	0.3	-24.9	-24.3	0.4	-21.1	14.5	4.4	12.8	3.6	0.4	300	0.749

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	20.7	14.5	18.8	13.2	17.4	12.5	15.0	19.6	13.9	17.8	12.9	16.2	3.5	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.5	9.0	16.4	11.9	8.7	15.3	11.2	8.3	14.4	41.7	19.7	38.8	18.0	36.5	16.3	20.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.2	10.6	8.8	-27.3	23.0	3.6	2.6	-29.9	24.8	-32.0	26.3	-34.0	27.8	-36.6	29.6		
			DB	WB	-27.5	16.0	3.5	1.2	-30.0	16.9	-32.1	17.7	-34.1	18.3	-36.7	19.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-5.2	-4.0	-3.1	2.0	6.8	10.6	12.6	11.5	8.4	3.2	-3.4	-3.8
	DBStd	8.16	8.26	6.43	5.87	3.17	2.50	2.08	2.05	1.86	2.54	4.13	6.93	6.97
	HDD10.0	2736	470	392	406	240	104	17	3	6	57	212	402	429
	HDD18.3	5592	728	625	664	490	357	232	179	211	297	471	652	687
	CDD10.0	185	0	0	0	0	4	35	82	53	9	1	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	12	0	0	0	0	0	0	8	3	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	3.5	3.7	4.0	3.4	3.7	3.6	3.3	3.3	3.2	3.4	3.5	3.4	3.8
Precipitation	PrecAvg	3808	341	322	286	302	267	176	218	286	434	412	391	401
	PrecMax	4552	911	598	561	654	626	293	394	577	925	863	1004	756
	PrecMin	2211	53	40	58	46	38	21	116	90	100	121	29	115
	PrecStd	522	188	156	140	144	173	66	76	132	197	164	257	161
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	7.3	8.8	12.5	18.8	21.7	24.6	22.9	17.2	12.2	8.2	7.2
		MCWB	5.5	3.3	4.3	7.2	10.9	14.8	16.9	15.5	11.8	9.0	5.7	4.9
	2%	DB	5.9	4.9	6.2	9.5	16.5	19.1	21.2	19.4	15.1	10.4	7.0	6.0
		MCWB	4.2	3.3	3.9	5.1	9.6	12.8	15.1	13.7	10.8	8.1	5.3	4.0
	5%	DB	4.8	4.0	4.9	7.8	14.6	17.7	19.1	17.7	13.5	9.2	6.0	4.9
		MCWB	3.4	2.5	3.1	4.1	8.6	12.4	14.0	12.9	10.3	7.1	4.5	3.2
	10%	DB	4.0	3.1	3.7	6.4	12.5	16.1	17.4	16.0	12.4	8.3	4.8	3.9
		MCWB	2.7	1.7	2.0	3.4	7.7	11.2	13.2	12.1	9.8	6.4	3.4	2.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	4.2	5.3	7.6	11.6	15.7	17.0	16.2	12.8	10.0	6.7	5.2
		MCDB	7.5	5.9	7.4	11.9	18.0	20.2	23.4	21.9	15.1	11.5	7.6	6.5
	2%	WB	4.6	3.4	4.2	5.6	10.0	13.8	15.3	14.4	11.7	8.7	5.6	4.3
		MCDB	5.7	4.9	5.9	8.6	15.4	18.4	20.4	18.6	13.9	10.2	6.8	5.8
	5%	WB	3.5	2.7	3.1	4.6	9.0	12.5	14.3	13.3	10.8	7.7	4.5	3.5
		MCDB	4.6	4.0	4.4	7.0	13.7	16.8	18.4	16.5	12.6	9.0	5.8	4.8
	10%	WB	2.7	1.9	2.2	3.9	8.2	11.5	13.4	12.4	10.3	6.7	3.5	2.8
		MCDB	3.8	2.9	3.4	5.9	11.9	15.4	16.8	15.1	12.0	8.0	4.6	3.8
Mean Daily Temperature Range	5% DB	MDBR	5.8	6.1	8.0	7.2	8.0	8.1	7.7	7.6	7.1	5.6	5.4	5.7
		MCDBR	4.2	4.2	6.4	10.6	14.7	13.8	13.3	13.3	10.3	6.1	4.5	4.3
		MCWBR	3.1	3.0	4.3	6.6	8.4	8.2	7.8	8.2	6.8	4.2	3.2	3.1
	5% WB	MCDBR	4.1	4.1	5.3	8.4	12.2	12.4	12.0	10.6	8.6	5.1	4.2	4.3
		MCWBR	3.0	3.0	3.6	5.5	7.2	7.6	7.2	6.9	6.0	3.9	3.2	3.3
Clear-Sky Solar Irradiance	taub	0.247	0.310	0.324	0.369	0.381	0.384	0.388	0.372	0.322	0.310	0.253	0.238	
	taud	2.053	2.064	2.076	2.015	2.116	2.246	2.307	2.352	2.478	2.274	2.042	1.912	
	Ebn at Noon	479	642	778	810	836	842	829	813	803	656	450	291	
	Edh at Noon	42	79	108	140	137	123	113	100	74	63	41	30	
All-Sky Solar Radiation	RadAvg	0.30	0.99	2.57	4.20	5.23	5.22	4.70	3.71	2.44	1.13	0.43	0.17	
	RadStd	0.04	0.15	0.30	0.43	0.45	0.28	0.37	0.41	0.35	0.14	0.07	0.02	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Regional (27 neighbors)	N/A	+1.90	+1.64	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page